

Qatar-1b

R-2729

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130905 · created 2026-09-02T00:13:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456540.82856 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.046
Transit depth (Rp/Rs)^2	1.02 +/- 0.93 [%]
Orbital Inclination (inc)	88.03 +/- 3.17
Airmass coefficient 1 (a1)	1.183 +/- 0.055
Airmass coefficient 2 (a2)	-0.097 +/- 0.036
Scatter in the residuals of the lightcurve fit is	4.68 %
Best Comparison Star	None
Optimal Aperture	3.31
Optimal Annulus	6.8
Transit Duration (day)	0.074 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.101 ± 0.046 vs 0.1463 (deviation 0.61σ)
Duration fit vs published	0.074 d vs 0.0692 d (deviation 0.27σ)
Mid-time O–C	0.2 min
Depth SNR	1.4
Residual scatter	4.68 %

Light curve

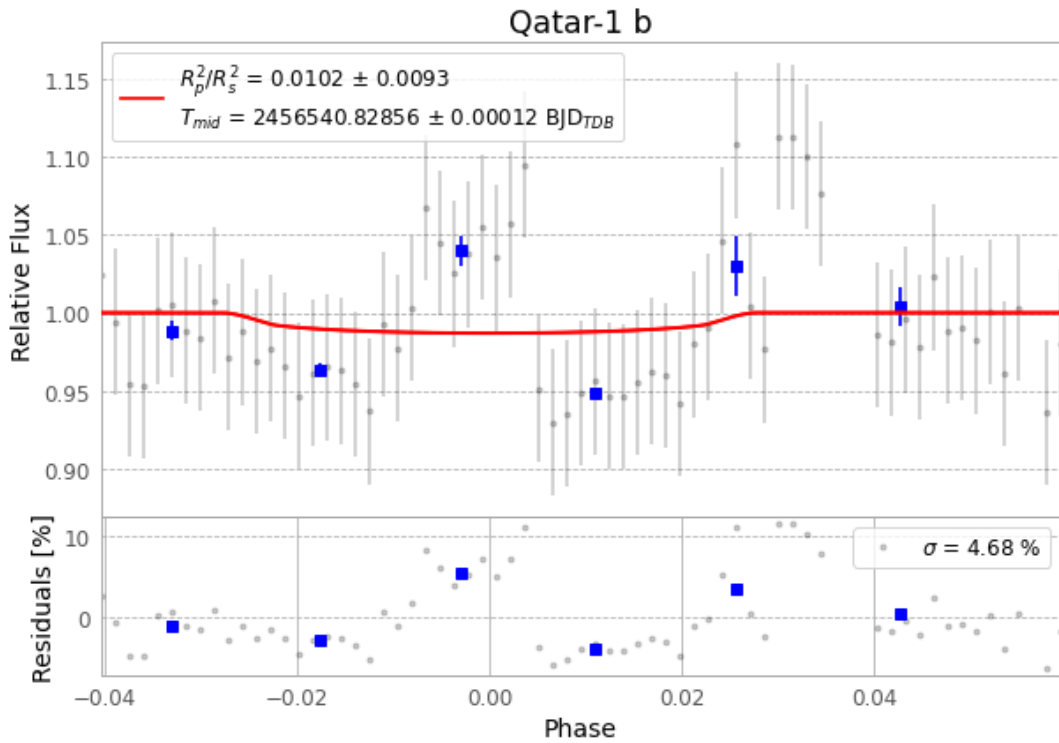


Figure 1 — FinalLightCurve_Qatar-1 b_2013-09-04.png (SHA-256 1ab1dde7100deadc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [411, 272], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7584feee8df7770e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), Qatar-1130905062713.FITS → Qatar-1130905095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130905.log (848d66bc28b0...), FinalLightCurve_Qatar-1 b_2013-09-04.png (1ab1dde7100d...), FinalLightCurve_Qatar-1 b_2013-09-04.csv (ffaff0491575...)

Compute resources

Wall time 187.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:13Z.